Exercise is associated with alterations to many basic physiological processes, including changes in metabolism and the cardiovascular system. As the intensity of exercise rises, both the cardiac output and oxygen consumption increase, as shown in Figure 1.

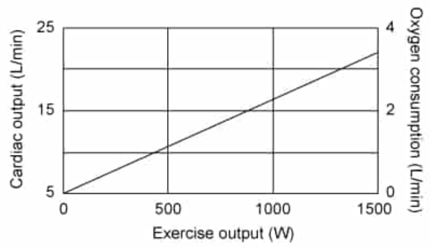


Figure 1 Changes in cardiac output and oxygen consumption in response to various levels of exercise

The cardiac output *CO* represents the volume of blood pumped by the heart in 1 minute. The *CO* is calculated from the heart rate *HR* and stroke volume *SV* using the equation

$$CO = HR \cdot SV$$

Equation 1

Experimental data has shown that the *HR* increases linearly with the exercise output E_{out}, leading to the relationship

$$HR = 0.65 \cdot E_{out} + 75$$

Equation 2

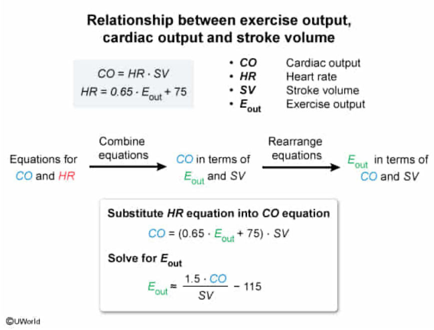
The metabolism of nutrients consumes oxygen and generates carbon dioxide as a waste product. The amount of carbon dioxide generated depends on the type of macronutrient that is consumed. For example, when carbohydrates are metabolized, every liter of oxygen consumed causes the generation of 1.0 liter of carbon dioxide. Alternatively, the metabolism of fats leads to 0.7 liters of carbon dioxide per liter of oxygen, and the metabolism of proteins generates 0.8 liters of carbon dioxide for each liter of oxygen.

What is the approximate relationship between E_{out}, *CO*, and *SV*?

- A. $E_{out} = \frac{0.65 \cdot CO}{SV} - 75$ [21%]
- B. $E_{out} = \frac{0.65 \cdot SV}{CO} - 75$ [15%]
- C. $E_{out} = 0.65 \cdot CO - 115 \cdot SV$ [4%]
- ✔ D. $E_{out} = \frac{1.5 \cdot CO}{SV} - 115$ [59%]

Correct
59% answered correctly

Explanation:



In this question, the three **variables** of interest (E_{out}, *CO*, and *SV*) are not provided in a single equation. The *CO* equation includes terms for *HR* and *SV* whereas the *HR* equation includes a term for E_{out}. However, both equations contain the same *HR* term, which can be **substituted** into the *CO* equation to generate a new equation that defines *CO* in terms of E_{out} and *SV*:

$$\text{CO} = \left(0.65 \cdot E_{\text{out}} + 75\right) \cdot \text{SV}$$

This new equation can be rearranged to solve for E_{out} in terms of *CO* and *SV*:

$$\frac{\text{CO}}{\text{SV}} = 0.65 \cdot E_{\text{out}} + 75$$

$$\frac{\text{CO}}{\text{SV}} - 75 = 0.65 \cdot E_{\text{out}}$$

The final form of this equation is:

$$E_{\text{out}} \approx \frac{1.5 \cdot \text{CO}}{\text{SV}} - 115$$

(Choice A) This expression results from a mathematical mistake when dividing by 0.65 to solve for E_{out}.

(Choice B) This equation results from mathematical mistakes in dividing by *SV* and 0.65 to solve for E_{out}.

(Choice C) This equation is obtained by incorrectly subtracting (75 · *SV*) from *CO* to solve for E_{out}.

Educational objective:
The equations for cardiac output and heart rate include the three variables E_{out}, *CO*, and *SV*. One equation can be substituted into the other, and the result rearranged to define the exercise output in terms of the cardiac output and stroke volume.

Time spent: 0 sec
Subject: Physics

QID:403264
System: Mechanics and Energy

Exercise is associated with alterations to many basic physiological processes, including changes in metabolism and the cardiovascular system. As the intensity of exercise rises, both the cardiac output and oxygen consumption increase, as shown in Figure 1.

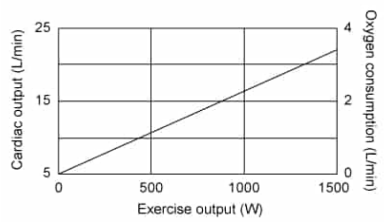


Figure 1 Changes in cardiac output and oxygen consumption in response to various levels of exercise

The cardiac output *CO* represents the volume of blood pumped by the heart in 1 minute. The *CO* is calculated from the heart rate *HR* and stroke volume *SV* using the equation

$$CO = HR \cdot SV$$

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Equation 2

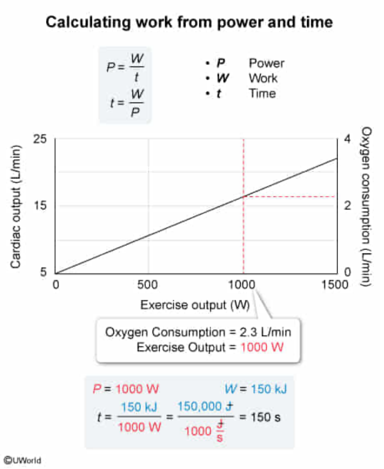
The metabolism of nutrients consumes oxygen and generates carbon dioxide as a waste product. The amount of carbon dioxide generated depends on the type of macronutrient that is consumed. For example, when carbohydrates are metabolized, every liter of oxygen consumed causes the generation of 1.0 liter of carbon dioxide. Alternatively, the metabolism of fats leads to 0.7 liters of carbon dioxide per liter of oxygen, and the metabolism of proteins generates 0.8 liters of carbon dioxide for each liter of oxygen.

How long does a person need to exercise to do 150 kJ of work when they consume oxygen at 2.3 L/min?

- A. 6.7 s [7%]
- B. 65 s [17%]
- ✔ C. 150 s [69%]
- D. 345 s [6%]

Correct
69% answered correctly

Explanation:



Mechanical work *W* is the energy put into an object when a force *F* is applied to the object over a distance *d*. The directions of *F* and *d* may be identical, or these directions may differ by an angle *θ*. The value of *W* depends on *F*, *d*, and *θ* according to the equation:

$$W = F \cdot d \cdot \cos\theta$$

Power *P* is an energy rate and can be defined as the ratio of *W* and time *t*:

$$P = \frac{W}{t}$$

In this question, the value of *W* done by a person is given as 150 kJ and their oxygen consumption is given as 2.3 L/min. From Figure 1, an oxygen consumption of 2.3 L/min represents a value of *P* equal to 1,000 W during the exercise. Hence, the above equation for *P* can be rearranged to calculate *t*:

$$t = \frac{W}{P}$$

$$t = \frac{150\ \text{kJ}}{1{,}000\ \text{W}}$$

Converting the units of *W* from kJ to J and the units of *P* from W to J/s yields:

$$t = \frac{150{,}000\ \cancel{\text{J}}}{1{,}000\ \frac{\cancel{\text{J}}}{\text{s}}} = 150\ \text{s}$$

Therefore, this person must exercise for 150 s with an output of 1,000 W to do 150 kJ of work.

(Choice A) 6.7 s results from incorrectly dividing *P* by *W* and leaving the units of *W* in kJ.

(Choice B) 65 s results from incorrectly leaving the units of *W* in kJ and dividing *W* by the person's oxygen consumption.

(Choice D) 345 s results from incorrectly multiplying the equation for *t* by the person's oxygen consumption.

Educational objective:

Power is the amount of work done per unit time. If the power generated during exercise is known, the time required to do a specific amount of work can be calculated.

Time spent:0 sec
Subject:
Physics

QID:403265
System:
Mechanics and Energy

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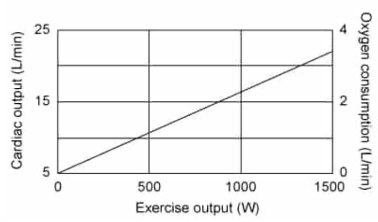

Figure 1 Changes in cardiac output and oxygen consumption in response to various levels of exercise

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Equation 1

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Equation 2

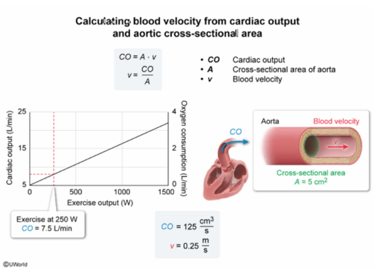
The metabolism of nutrients consumes oxygen and generates carbon dioxide as a waste product. The amount of carbon dioxide generated depends on the type of macronutrient that is consumed. For example, when carbohydrates are metabolized, every liter of oxygen consumed causes the generation of 1.0 liter of carbon dioxide. Alternatively, the metabolism of fats leads to 0.7 liters of carbon dioxide per liter of oxygen, and the metabolism of proteins generates 0.8 liters of carbon dioxide for each liter of oxygen.

A runner's exercise output is 250 W. If the cross-sectional area of their aorta is 5 cm^2, what is the average velocity of the blood flowing through their aorta?

- A. 0.25 m/s [19%]
- B. 1.5 m/s [33%]
- C. 2.5 m/s [39%]
- D. 15 m/s [7%]

Correct
19% answered correctly

Explanation:



The **cardiac output** *CO* represents the volumetric flow rate of blood exiting the left ventricle and entering the aorta. The *CO* is the product of the cross-sectional area *A* of the aorta and the velocity *v* of the blood:

$$\text{CO} = A \cdot v$$

This equation can be rearranged to solve for *v*:

$$v = \frac{\text{CO}}{A}$$

In this question, *A* is given as 5 cm^2 and the exercise output of the runner is given as 250 W. From Figure 1, an exercise output of 250 W equates to a *CO* value of approximately 7.5 L/min. Converting the units of *CO* from L/min to cm^3/s yields:

$$\text{CO} = \left(7.5\,\frac{\cancel{\text{L}}}{\cancel{\text{min}}}\right)\left(1{,}000\,\frac{\text{cm}^3}{\cancel{\text{L}}}\right)\left(\frac{1}{60}\,\frac{\cancel{\text{min}}}{\text{s}}\right) = 125\,\frac{\text{cm}^3}{\text{s}}$$

Substituting the values of *CO* and *A* into the equation for *v* gives:

$$v = \frac{125\ \frac{\text{cm}^{\cancel{3}}}{\text{s}}}{5\ \cancel{\text{cm}^2}} = 25\ \frac{\text{cm}}{\text{s}} = 0.25\ \frac{\text{m}}{\text{s}}$$

(Choice B) Calculating the answer without correctly converting the units of *CO* gives the value of 1.5 m/s.

(Choice C) Calculating the answer using a factor of 10 instead of the correct factor of 100 when converting *v* from cm/s to m/s gives the value of 2.5 m/s.

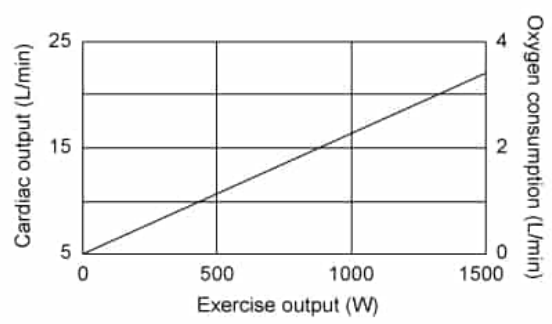
(Choice D) Calculating the answer without correctly converting *v* to m/s gives the value of 15 m/s.

Educational objective:
The velocity of blood in the aorta is equal to the ratio of the cardiac output and the cross-sectional area of the aorta.

Time spent:0 sec
Subject:
Physics

QID:403266
System:
Mechanics and Energy

Exercise is associated with alterations to many basic physiological processes, including changes in metabolism and the cardiovascular system. As the intensity of exercise rises, both the cardiac output and oxygen consumption increase, as shown in Figure 1.



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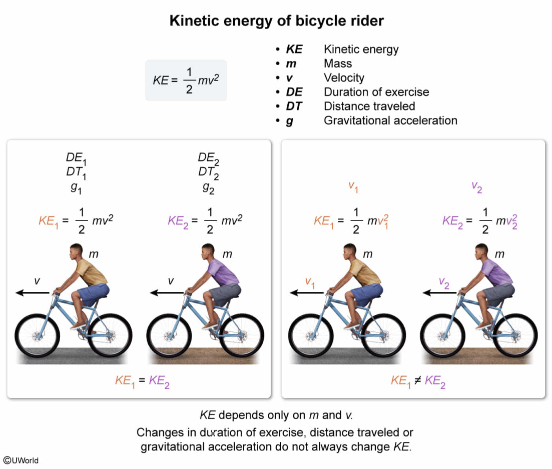
Changing which of the following will always change the kinetic energy of a bicycle rider?

- A. Duration of exercise
- B. Velocity [95%]
- C. Distance traveled [1%]
- D. Gravitational acceleration [2%]

Correct

95% answered correctly

Explanation:



The kinetic energy *KE* of an object depends on the **mass** *m* and **velocity** *v* of the object, based on the equation:

$$\text{KE} = \frac{1}{2}mv^2$$

In this question, changes in the *KE* of a bicycle rider are considered when the duration of exercise, the velocity, the distance traveled, or the gravitational acceleration are changed. From the equation above, the *KE* depends only on *m* and *v*. Therefore, only changes in *m* or *v* will affect the *KE*.

(Choices A, C, and D) The duration of exercise, distance traveled, or gravitational acceleration do not necessarily influence the *KE* of the bicycle rider. For example, all these parameters could change but the *KE* of the rider will remain unchanged if traveling along a horizontal road at constant *v*.

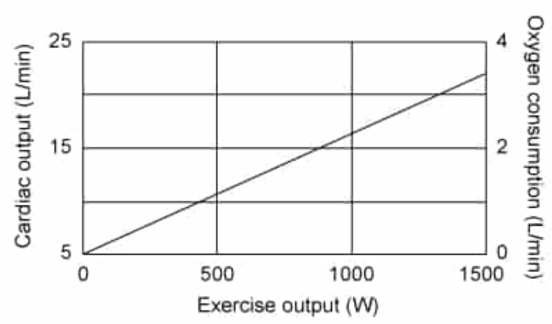
Educational objective:

The kinetic energy of an object depends only the object's mass and velocity. Other parameters, such as the duration of motion, distance traveled, or gravitational acceleration, do not affect the kinetic energy of an object.

Time spent:0 sec
Subject:
Physics

QID:403267
System:
Mechanics and Energy

Exercise is associated with alterations to many basic physiological processes, including changes in metabolism and the cardiovascular system. As the intensity of exercise rises, both the cardiac output and oxygen consumption increase, as shown in Figure 1.



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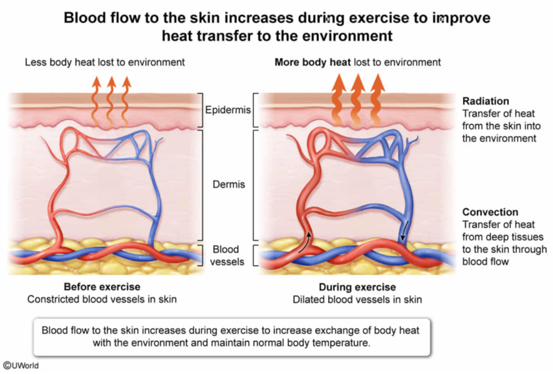
Does blood flow to the skin increase during exercise?

- A. Yes, because regulation of body temperature is improved [49%]
- B. Yes, because blood supply to skeletal muscles is improved [47%]
- C. No, because blood supply to skeletal muscles is reduced [2%]
- D. No, because cardiac output is reduced

Correct

49% answered correctly

Explanation:



Exercise generates additional body heat that must be released into the environment to maintain **body temperature** within the normal range. Some ways in which the body exchanges excess heat with its surroundings include sweating, increased respiration, and vasodilation of blood vessels in the skin. These physiological responses dissipate heat into the environment through a variety of heat transfer mechanisms, including **radiation**, **convection**, and **conduction**.

Vasodilation of blood vessels in the skin causes increased blood flow to the surface of the body. The increased blood flow transports excess heat from deep inside the body to the surface. This achieves heat transfer through convection because the movement of blood is employed to transport heat from one part of the body to another.

Once blood flow transports excess heat to the surface of the body, the heat can be exchanged with the environment more effectively. This transfer of heat occurs through radiation in the form of electromagnetic waves in the infrared region of the spectrum emitted from the skin into the environment.

In this question, blood flow to the skin increases during exercise to increase the transfer of excess body heat to the environment and maintain normal body temperature.

(Choice B) Increasing blood flow to the skin does not increase blood flow to the skeletal muscles.

(Choice C) Increasing blood flow to the skin may have the detrimental effect of decreasing blood flow to the skeletal muscles. However, the need to maintain normal body temperature is more important than maximizing muscle blood flow.

(Choice D) Increasing blood flow to the skin does not decrease the cardiac output.

Educational objective:
During exercise, blood flow to the skin increases to maintain normal body temperature. Excess body heat generated during exercise can be transferred to the surface through convection and dissipated into the environment through radiation.

Time spent:0 sec | **QID:**403268

Subject: Physics | **System:** Mechanics and Energy

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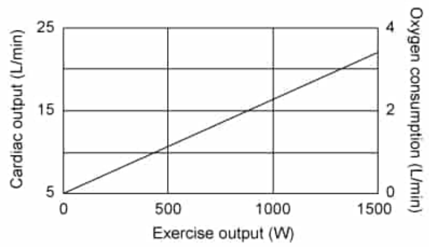

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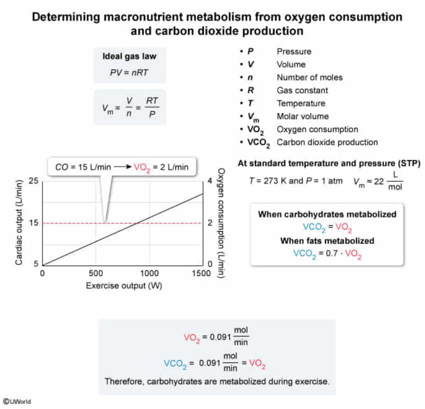
While exercising on a treadmill, a person has a cardiac output of 15 L/min and produces carbon dioxide at 0.091 mol/min. What is their ratio of metabolizing carbohydrates to fats during the exercise? (Assume all gasses are ideal and are at STP.)

- A. 1:2 [21%]
- B. 1:1 [16%]
- ✔ C. 1:0 [16%]
- D. 2:1 [44%]

Correct
16% answered correctly

Explanation:



An ideal gas follows the equation that the product of pressure *P* and volume *V* equals the product of the number of moles *n*, the gas constant *R*, and the temperature *T*:

$$PV = nRT$$

Standard temperature and pressure (STP) refers to the situation where *T* = 273 K and *P* = 1 atm. Under STP, an ideal gas has the same ratio of *V* and *n*, known as the **molar volume** V_m. Rearranging the ideal gas law to solve for V_m yields:

$$V_m = \frac{V}{n} = \frac{RT}{P} = \frac{\left(0.082\frac{L \cdot \cancel{atm}}{\cancel{K} \cdot mol}\right)(273\ \cancel{K})}{1\ \cancel{atm}} \approx 22\ \frac{L}{mol}$$

According to the passage, metabolizing carbohydrates generates 1 liter of carbon dioxide for each liter of oxygen consumed. However, the metabolism of fats generates 0.7 liters of carbon dioxide per liter of oxygen consumed.

In this question, the person's cardiac output is 15 L/min, which corresponds to an oxygen consumption VO_2 of 2 L/min based on Figure 1. At STP, the VO_2 in mol/min is:

$$VO_2 = \left(2\ \frac{\cancel{L}}{min}\right)\left(\frac{1}{22}\ \frac{mol}{\cancel{L}}\right) = 0.091\ \frac{mol}{min}$$

Therefore, the VO_2 and carbon dioxide production VCO_2 are equal, indicating that this person is metabolizing only carbohydrates:

$$VO_2 = VCO_2 = 0.091\ \frac{mol}{min}$$

(Choice A) Metabolizing carbohydrates and fats at a 1:2 ratio would produce 0.8 L of carbon dioxide for each liter of oxygen consumed and result in a VCO_2 of 0.073 mol/min.

(Choice B) Metabolizing carbohydrates and fats at a 1:1 ratio would produce 0.85 L of carbon dioxide for each liter of oxygen consumed and result in a VCO_2 of 0.077 mol/min.

(Choice D) Metabolizing carbohydrates and fats at a 2:1 ratio would produce 0.9 L of carbon dioxide for each liter of oxygen consumed and would result in a VCO_2 of 0.082 mol/min.

Educational objective:
The ideal gas law implies that the molar volume is 22 L/mol for all gases at standard temperature and pressure. Measuring the oxygen consumption and carbon dioxide production during exercise can identify which macronutrients are metabolized.

Time spent:0 sec
Subject:
Physics

QID:403269
System:
Mechanics and Energy